MAC21 — The 48-Hour Plan

Now: Sunday 12:00 · **Exam:** Tuesday 09:30 · **Real study time available:** ~22 hours · **Target:** 60–70 marks by the easiest route

The whole strategy in one line

In each unit, answer **a) + b) + ONE of the two 7-markers**. That is `2 + 4 + 7 = 13 marks per unit × 5 units = **65 marks**`.

You do not need both 7-markers. That halves the hardest part of the syllabus and is why this is achievable in two days. Anything you manage on the second 7-marker is bonus on top of 65.

The 5 questions you are *likely* answering

Decide in the hall, not now. The question numbers below are where these topics have *usually* sat — they are a tiebreaker, not a commitment. Topics do not vanish when a paper is reshuffled, they just move. If "expand about (1, π/2)" turns up in Q1 instead of Q2, you answer Q1. **Study the topics; choose the question after reading both options.**

UNIT	ANSWER	A) — 2 MK	B) — 4 MK	YOUR ONE 7-MARKER	TOTAL
I	**Q2**	State Maclaurin's / Taylor's series	`xy(1−x−y)` max at `(1/3,1/3)`	Expand `xy+cos(xy)` about `(1, π/2)` **4×**	**13**
II	**Q4**	DE of L–C circuit / define orthogonal trajectories	Bungee Euler *or* Taylor's method *or* orthogonal trajectory	Modified Euler's method **5×** *(sits at **4c** in 4 of 5)*	**13**
III	**Q5**	Steps for solving Cauchy's LDE	Prove `1/(D+a)·X` **4×**	Cauchy's equation **4×** *(easier than variation of parameters)*	**13**
IV	**Q8**	Lagrange's inverse interpolation formula	Find the missing terms **3×**	Newton's divided difference **5×** *or* Simpson's 3/8	**13**
V	**Q10**	Rank / echelon / consistency definition	Gauss-Seidel, 2 iterations **3×**	Gauss elimination word problem *(easier than system of ODEs)*	**13**
				Floor if every 7-marker lands	**65**

What you are deliberately NOT studying. Every one of these lives in the question you are *not* answering. Touch them only if you finish everything else:
Legendre's equation (Q6) · **variation of parameters** (the other 7 in Q5 — learn only if time) · **Rayleigh's power method** (Q9) · **diagonalization / modal matrix** (Q9) · **Lagrange multipliers** (Q1) · **maxima-minima beyond `xy(1−x−y)`** (Q1) · **radius of curvature, Newton-Cotes derivation, backward difference tables**.
That is roughly 40% of the syllabus, cut on purpose.

How exposed are you really? — syllabus coverage audit

This checks the plan against **your 28 sessions**, not against the predictions. It is the honest answer to "what if the paper ignores the prediction sheet".

UNIT	SESSIONS	PLAN ALONE	+ INSURANCE	WHERE THE EXPOSURE IS
II	5	**5 / 5**	**5 / 5**	**Zero exposure — and this is arithmetic, not a prediction.** Unit II has 5 sessions and four of them *are* the numerical methods. No question can be set that avoids them.
I	6	**4 / 6**	**6 / 6**	Missing Lagrange multipliers and 1-variable expansion. Both cheap to add.
III	6	**4 / 6**	**6 / 6**	Cauchy *forces* you to learn CF + P.I. anyway, so sessions 12–14 come free. Missing Legendre and variation of parameters — and Legendre is 20 minutes once Cauchy is in.

IV	6	4 / 6	6 / 6	Missing numerical differentiation and Newton-Gregory as a full problem. Both are formula-application.
V	5	2 / 5	5 / 5	**This is the real hole.** No eigenvalues, so no diagonalization, no Rayleigh, no system of ODEs. If both Q9 and Q10 lean eigen-heavy you are short.
Total		~19 / 28	28 / 28	Plan alone ≈ **68%** of the syllabus. With the insurance block below, effectively all of it.

The 90 minutes that removes most of the risk

Eigenvalues and eigenvectors. Sessions 26, 27 and 28 are *one chain* — diagonalization, Rayleigh's power method and system-of-ODEs-by-matrix are all "find the eigenvalues, then do one more step". Learn the eigen-step once and all three unlock together.

One 90-minute block takes Unit V from 2/5 to 5/5. It is the highest-leverage block in the whole two days.

SUNDAY — 12:00 to 23:00 (~8.5 h study)

Today you bank **Units III, IV and V** — the three most predictable units, worth 39 marks. Do the hardest-to-fake unit while you are freshest.

TIME	BLOCK	WHAT EXACTLY
12:00–13:00	**2-mark bank, pass 1**	Section 6 of the prediction sheet. Write each definition out *by hand*, once. Do not just read. This is 10 marks and it is the cheapest hour of the whole weekend.
13:00–13:45	*Lunch*	*Away from the desk.*
13:45–16:00	**UNIT III** 13 mk	The single biggest win in the paper. • Memorise the `1/(D+a)` proof cold — it is 4 marks, appeared **4×**, and never changes. • Cauchy's equation: the substitution $x = e^z$, then it is a constant-coefficient problem. • Work M23 **5c**, M24 **5d**, MAM24 **5d** — three Cauchy problems. • Write the "steps for solving Cauchy's LDE" answer once.
16:00–16:20	*Break*	*Walk. No screen.*
16:20–18:30	**UNIT V** 13 mk	Smallest unit in the paper. • Gauss-Seidel — do the MP/P26 system (`x+8y+2z=-4` ...), it appeared **twice identically**. • Then M23 **9b** and M24 **10b**. Three reps and it is automatic. • Gauss elimination: M23's electrical network and M24's dietician problem. • The five Unit V definitions (rank, echelon, consistency, modal & spectral, "is every matrix diagonalizable").
18:30–19:30	*Dinner*	*Proper meal.*
19:30–21:30	**UNIT IV** 13 mk	The most mechanical unit — good for a tired brain. • Missing terms — do the `45…65 / 3.0, –, 2.0, –, -2.4` table, it appeared **identically twice**. • Newton's divided difference — the `(2,4)(4,56)(9,711)(10,980)` cubic appeared **twice verbatim**. Do it. • Simpson's 3/8 on $\int_0^1 dx/(1+x^2)$ as your backup 7-marker. • Lagrange's inverse interpolation formula — write it out.
21:30–22:30	**Consolidate**	Close the notes. From memory: the `1/(D+a)` proof, one Cauchy problem, one Gauss-Seidel. Whatever you cannot do, mark it and fix it first thing tomorrow.
22:30–23:00	*Wind down*	*No maths. Sleep by 23:00.*

End of Sunday you should own: Units III, IV, V a+b+one 7 — **39 marks** — plus most of the 2-mark bank.

MONDAY — 07:00 to 23:00 (~11.5 h study)

Units I and II in the morning, a timed mock at midday, repairs and bonus marks in the evening.

TIME	BLOCK	WHAT EXACTLY

07:00–07:30	*Wake, breakfast*	—
07:30–09:45	**UNIT II** 13 mk	Learn the four numerical methods *as one family* — they are variations of a single idea, and doing them together is far faster than one at a time. • Taylor's series method → Euler's → **Modified Euler's** → RK4. Same skeleton: tabulate, iterate, 4 decimals. • **Modified Euler is your 7-marker** (slot **4c**, 4 of 5 papers). Do three of them. • Newton's law of cooling — 10 minutes to learn, appeared **5×**. Do the coffee problem. • Orthogonal trajectories, Cartesian and polar. The circuit-DE 2-marker.
09:45–10:00	*Break*	—
10:00–12:00	**UNIT I** 13 mk	• **Expand about** $(1, \pi/2)$ — appeared **4×**, the point never changes. Do all three variants: $xy+\cos(xy)$, $xy+\sin(xy)$, $xy^2+\cos(xy)$. • $xy(1-x-y)$ maximum at $(1/3, 1/3)$ — appeared **3×**. Learn the r-s-t test on it. • The Newton-Raphson system $x^2+y^2=x,\ x^2-y^2=y$ from $(0.8,\ 0.4)$ — appeared **4×**. It is in Q1, but learn it: it is your escape hatch if Q2 looks bad. • The Unit I statements: Maclaurin one variable, Taylor two variables.
12:00–13:00	*Lunch*	—
13:00–15:00	**TIMED MOCK** critical	Sit the **Model Paper**. Answer **Q2 · Q4 · Q5 · Q8 · Q10**, a + b + one 7-marker each. Clock it: **2 hours**. Closed book, no phone. This is the most important block of the two days — it converts "I know it" into "I can write it under time".
15:00–15:30	*Break*	*Mark it honestly first, then rest.*
15:30–17:30	**Repair**	Whatever you dropped in the mock, redo — the actual question, start to finish, not just "review the method". Anything you got right, do not touch again.
17:30–18:30	*Dinner*	—
18:30–21:00	**INSURANCE** do not skip	This block is what makes the plan robust to a reshuffled paper. Strictly in this order — each item is ranked by how much syllabus it unlocks per minute: • **Eigenvalues & eigenvectors — 90 min.** Unlocks diagonalization, Rayleigh's power method *and* system of ODEs by matrix method in one go. Takes Unit V from 2/5 to 5/5. **Highest value block of the weekend.** • **Legendre's equation — 20 min.** Same substitution as Cauchy with $(ax+b) = e^z$. Nearly free once Cauchy is in, and it closes Q6. • **IVP / BVP — 15 min.** Free once you have CF + P.I. Just apply two conditions. • **Variation of parameters — 45 min** if time. Appeared **5×/5**; it is the other 7-marker in your own Q5.
21:00–22:15	**2-mark bank, pass 2**	Write out all five units' definitions from memory. Then every formula you need: N-R, Modified Euler, RK4, Newton-Gregory, Lagrange, Simpson's 1/3 and 3/8, Gauss-Seidel, Cauchy substitution. One sheet, by hand.
22:15–23:00	*Wind down*	***Sleep by 23:00.*** *Non-negotiable — see below.*

TUESDAY — 06:00 to 09:30

TIME	BLOCK	WHAT EXACTLY
06:00–06:40	*Wake, breakfast*	*Eat properly. Do not skip it.*
06:40–07:40	**2-mark bank, final**	All 5 units' definitions + your formula sheet, from memory. This is 10 marks and it is the last thing that should still be wobbling.
07:40–08:30	**Setups only**	For each of your five 7-markers, write only the *first three lines* — the setup, not the full solve. $1/(D+a)$ proof · Cauchy substitution · Modified Euler table · divided-difference table · Gauss-Seidel rearrangement. Ten minutes each.
08:30–08:50	**Skim**	Tier 1 table of the prediction sheet. Read, do not solve.
08:50	***STOP***	*Close everything. Travel.* ***Learn nothing new after this point*** *— it displaces what is already in place and costs more than it adds.*

In the exam hall — 09:30 to 12:30

MINUTES	DO THIS

0–5	**Read all ten questions before writing anything.** In each unit compare the two options and commit. Write your five choices on the top of the answer sheet so you do not drift.
5–20	**All five 2-markers first.** 10 marks in 15 minutes, while you are calm. Nothing else in the paper has that rate of return.
20–50	**All five 4-markers.** Another 20 marks. You are now at 30 with 100 minutes left.
50–170	**The 7-markers — easiest first.** 24 minutes each × 5 = 120. Do your strongest unit first to build momentum. If one stalls past 25 minutes, leave it and move on; come back only if there is time.
170–180	**Sweep.** Check every question is numbered and lettered correctly. Attempt the second 7-marker anywhere you have a spare minute — partial marks are still marks.

Show every step. These are method-marked papers. A Gauss-Seidel table with an arithmetic slip in the last row still scores most of the 4. A correct final answer with no working does not. **Never leave a part blank** — write the formula and the setup even if you cannot finish; that is 2–3 marks on a 7-marker for ninety seconds of work.

Sleep 7 hours both nights. This is the highest-leverage instruction on the page. Studying until 2am costs you more in recall and arithmetic accuracy than the extra hours add — and this paper is almost entirely arithmetic accuracy. If you are behind on Monday night, cut the bonus 7-markers, not the sleep.

If you open the paper and it looks unfamiliar anyway. Do not panic-read all ten questions twice — work the structure instead.
1. The five 2-markers are drawn from a small fixed pool of definitions. Take them first regardless of how the rest looks. That is **10 marks** banked before you have solved anything.
2. In each unit, count how many parts of Q*odd* vs Q*even* you can start. Pick on the count, not on which looks nicer.
3. These are method-marked. On any 7-marker you cannot finish, write the formula, the setup, and the first two steps — that is routinely 3 of 7 for ninety seconds' work. **Five partial attempts beat two perfect answers and three blanks.**
4. A "new" question is usually a familiar method in unfamiliar clothing. Radius of curvature was an interpolation problem. A dietician's meal plan was Gauss elimination. Ask "what data am I given?" before "what topic is this?"

If you fall behind — the order to sacrifice things:
1. Variation of parameters (last item of the insurance block) — drop it, the first three items matter far more.
2. Unit I depth — keep only "expand about (1, π/2)" and `xy(1−x−y)`.
3. Simpson's 3/8 backup in Unit IV — divided difference alone is enough.
Never sacrifice: the 2-mark bank · the `1/(D+a)` proof · Gauss-Seidel · missing terms · Modified Euler · **and the eigenvalue block**. Those are ~25 marks plus your entire Unit V safety net, for about five and a half hours of work.

Built from the MAC21 prediction sheet: 6 papers analysed, repeat counts stated out of the papers carrying each unit. Question-choice recommendations reflect observed slot frequency, not certainty — always read both options before committing. Timings assume you start at Sunday 12:00 and are asleep by 23:00 both nights.